ehbea

2013

Index

Welcome

Welcome to the 8th Annual Conference of the European Human Behaviour and Evolution Association at VU University Amsterdam!

We are delighted to welcome everyone to Amsterdam for the 8th Annual EHBEA conference. We have a wonderful program this year in a wonderful city. Amsterdam is home to many world class tourist attractions such as the Van Gogh Museum, the Artis Zoo, the Anne Frank House, and our famous canals, which are a UNESCO World Heritage Site. In addition to the many intellectual treasures to be found at the conference, we hope you will take pleasure in visiting our beautiful city too.

This year we welcome a record number of around 220 attendees to our conference. For the scientific program, we selected 43 presentations and 95 posters from a record number of submissions covering a wealth of topics in evolution and human behavior, representing the diversity, breadth, and depth of our fast-growing field.

In addition there are several key note presentations by eminent scientists covering different research areas of evolution and human behavior. We are delighted to welcome our five key note speakers Bram Buunk, Simon Gaechter, Kristen Hawkes, Joe Henrich, and Cecilia Heyes to Amsterdam. This year's conference opens on Sunday afternoon 24 March at 4pm with a key note by Joe Henrich. Thereafter there will be the welcome reception. Both are events you do not want to miss!

On Monday morning we are delighted to welcome the Rector of the University, Lex Bouter, who will officially open the conference.

After a full day of talks there will be a round table panel discussion in the early evening on Open Access publishing, titled *"The Future of Publishing in Evolutionary Sciences."* This discussion is sponsored by the Dutch Science Foundation, NWO. With (hopefully) some audience participation this promises to be a lively event.

Later in the evening on Monday there will be – for those who have registered -- a boat ride visiting the many beautiful canals and waterways in Amsterdam. After another full day of talks on Tuesday 26 March there will be the traditional poster session accompanied by drinks and snacks in the early evening. We have many excellent poster submissions this year so this is an event you should attend.

On Wednesday 27 March the Young Investigator Award of EHBEA will be given to the recipient David Lawson. In the evening there will be the annual EHBEA Assembly General Meeting followed by an optional diner (again for those who have registered) in the beautiful Hermitage Museum in the centre of Amsterdam.

We are grateful for the enormous time and effort so many people devoted to organizing this year's conference. We thank the members of the Program Committee who took on the unenviable and often heart-breaking task of selecting this year's presentations from the sea of excellent submissions. In addition, the following VU University graduate students deserve special thanks for their assistance with the conference: Nancy Blaker, Allen Grabo , Jannie Kaatee, Jill Knapen, Michael Laakasuo, Zoi Manesi. We would also like to thank Marco Benard from VU University ICT for developing the EHBEA2013 website (http://www.ehbea2013.com/) and Stefan de Graaf from the Department of Social & Organizational Psychology for sorting out travel and accommodation.

Finally, we would like to acknowledge the following sponsors of the conference whose financial support is greatly appreciated: NWO, the Galton Institute, Springer, and the Department of Social & Organizational Psychology and the Department of Sociology at VU University Amsterdam.

We hope you enjoy everything the conference and the city of Amsterdam has to offer this year!

The Organizing Committee of EHBEA 2013,

Mark van Vugt

Thomas Pollet

Fleur Thomese

Josh Tybur

Abstracts: Keynote speakers

Culture-Gene coevolution and the origins of human cooperation

Joe Henrich (henrich@psych.ubc.ca)
Canada Research Chair in Culture, Cognition and Coevolution, Department of Psychology, Department of Economics, University of British Columbia

Our species is unique in being both highly cultural, heavily reliant on acquiring essential aspects of our behavioral repertoire from others, and ultra-social, capable of cooperating on large-scales with non-relatives. Research over the last thirty years examining the interaction between genetic and cultural evolution suggests that these two features—culture and cooperation—are deeply intertwined. While the consideration of such culture-gene coevolutionary processes are taken seriously for understanding many aspects of the human phenotype, and represent the leading view in some cases, the origins of human cooperation, status and ultra-sociality remain a flashpoint. Drawing on evolutionary modeling, cross-cultural and cross-species experiments, laboratory studies of social learning, and quantitative ethnographic work on social life in small-scale societies, I will argue that cultural evolution, driven by competition among social groups, has shaped human genetic evolution and the emergence of our unique social psychology. This approach allows us to tackle otherwise puzzling aspects of human cooperation, institutions, religions, marriage systems, and social norms, as well as giving us a purchase to grapple with the broadest patterns in human history.

Learning to read minds

Cecilia Heyes (cecilia.heyes@all-souls.ox.ac.uk)
All Souls College & Department of Experimental Psychology, University of Oxford

Even researchers who emphasize the power of cultural evolution typically assume that ‘cultural learning’, the cognitive mechanisms underwriting cultural evolution, are genetically inherited. In contrast, I will argue that it is not just the grist but also the mills that are cultural in origin; humans learn from others not just facts about the world and skills for dealing with it (grist), but also the cognitive processes that make ‘fact inheritance’ possible (mills). Literacy is a paradigmatic example of the cultural inheritance of cultural learning. Previously I have argued that social learning, imitation, and mirror neurons provide further examples. In this talk, I focus on mentalizing (also known as ‘theory of mind’, ‘folk psychology’ and ‘mindreading’). First, to clear the ground, I will examine experimental work that has recently re-invigorated the nativist view of mentalizing. This work seems to show that infants and adults automatically represent what others see, intend and believe; that they engage in ‘implicit mentalizing’. I will argue that these results are due to ‘sub-mentalizing’ - domain-general cognitive processes that can simulate mentalizing in social contexts. In the second part of the paper, I will suggest that learning to mentalize is a lot like learning to read. Both acquisition processes are typically guided by expert instruction, and involve the reconfiguration of cognitive parts into a new system. In the case of reading, the crucial parts include generic object recognition processes, attentional routines, and grapheme-phoneme correspondence rules. In the case of mindreading, they include generic inference processes, attentional routines, and behaviour rules.

Jealousy and intrasexual competition: From unconscious processes to hormonal influences

Abraham P. Buunk (a.p.buunk@rug.nl)
Evolutionary Social Psychology, University of Groningen and Royal Netherlands Academy of Arts and Sciences

Among human males, jealousy has a long evolutionary history rooted in fights over access to females and mate guarding. Among human females, jealousy seems to have evolved to obtain and preserve the investment of males in offspring. Jealousy implies by definition a rival, and assessing the threat of a rival is a basic, adaptive mechanism, that is directly related to intrasexual competition. I will present an overview of our research on this issue in the past decade. I will discuss sex differences and similarities in the rival characteristics that evoke jealousy. I will show that the sex differences are similar among gay men and lesbian women. I will also show that, while females are overall more than males concerned about the physical attractiveness of the rival, when it concerns sexual infidelity males are more concerned about the physical attractiveness of the rival. I will show that individuals can unconsciously assess the threat of a rival when they are subliminally exposed to rivals. While there does not seem a relationship between testosterone and jealousy, less masculine men, i.e., men with a high 2d:4D are more jealous of physically dominant rivals, and less feminine women, i.e. with a low 2d:4D more jealous of physically attractive rivals. Overall, men are more jealous as they are shorter, especially of dominant rivals. Women are more jealous and more intrasexually competitive when they are either relatively short or relatively tall. I will discuss recent research from our group that can explain these sex differences in height. Finally, I will show that jealousy and intrasexual competition among women are elevated when they are fertile. In general, our research illustrates the strong biological and evolutionary basis of jealousy.

Understanding cooperation: A behavioural economics perspective

Simon Gächter (simon.gaechter@nottingham.ac.uk)
School of Economics, University of Nottingham & KNAW, VU University Amsterdam

Understanding cooperation – collectively beneficial collaboration that is not in an individual's self-interest – is an important challenge across the behavioural sciences. In this talk I will present some insights from a behavioural economics perspective. I will introduce the workhorse of most recent experiments- the public goods game. I will show that many people cooperate against their self-interest, but that cooperation is inherently fragile. The reason is that many people are 'conditional cooperators' who only contribute if others contribute. I will then present some evidence on the power of punishment to sustain cooperation. Many people are willing to punish free loaders, and this attitude is supported by moral judgments and negative emotions about free riders. In the last part of my talk I will address challenges to the lab results: the generalisability of results from students to other people; the external validity problem; the costs of punishment, which might make its evolution hard to explain; and the role of anonymity, which might exaggerate punishment. I will show that under some fairly minimal conditions full cooperation can be sustained with no punishment at all.

Grandmothering and human evolution

Kristen Hawkes (hawkes@anthro.utah.edu)
Department of Anthropology, University of Utah

Some of the living great apes are more closely related to humans than they are to each other. But the gulf between all of them and us seems enormous. Why are humans so different? One thing that distinguishes us from them is our extraordinary longevity. I¹ll review a Grandmother Hypothesis that might help explain why such long lifespans together with other striking life history features evolved in our lineage, then consider why grandmothering may be the foundation for features of sociality that are distinctively human.

Abstract: EHBEA New Investigator Award Winner

Natural selection on wealth and fertility in humans

David W. Lawson (d.lawson@ucl.ac.uk)
Department of Anthropology, University College London

Life history theory argues that all organisms have two main goals – 1) competitively extract resources from the ecological and social environment and then 2) selectively allocate these resources to reproduction in a way that maximizes inclusive fitness. In this talk, building on the work of a number of evolutionary anthropologists and demographers, I argue that natural selection has shaped humans to rely on largely distinct proximate pathways to address these two goals. Resource accumulation is a cognitively taxing and complex social process, requiring considerable context-dependent plasticity and conscious goal-directed strategizing. A review of studies of fertility and offspring success however suggests that automatic physiological mechanisms, which suppress reproduction only when maternal or child survival is at immediate risk, have been sufficient to ensure optimal reproductive behaviour throughout most of human history. Recognizing this distinction exposes our inherent vulnerability to maladaptive decision-making in novel environments where wealth accumulation is now in conflict with reproductive opportunities for both men and women – improving our understanding of why fertility rates plummet when populations undergo socioeconomic development (i.e. the demographic transition). I review empirical evidence that supports the hypothesis that modern low fertility rates can be understood as a response to increased status competition and returns to high parental investment, and that fertility limitation is ultimately maladaptive in terms of both short and long-term fitness. Finally, I critique recent studies concluding that natural selection continues to act positively on wealth even in contemporary low fertility populations, and argue that natural selection is now acting negatively on strategies of wealth accumulation and is for the first time strongly favouring individuals that desire early childbearing and large families regardless of the socioeconomic consequences.

Abstracts: Talks
(presenter/corresponding author is underlined)

Monday, 25th: 9:55–10:20, Aula
Cooperative behaviour as a signal in partner choice

Gilbert Roberts (gilbert.roberts@ncl.ac.uk)
Newcastle University

Objective
In theory, being seen to be cooperative could pay if the costs of investing in reputation-building displays are outweighed by the benefits of attracting a cooperative partner. But how could such a signalling system be honest? One proposal is via the handicap principle or costly signalling, yet there remain questions as to why this mechanism would result in cooperative signals. My objective here was to determine when investment in cooperative reputations can pay and what is the mechanism underlying honest signalling.

Methods
I developed a model (based on a framework referred to in the literature as competitive altruism) comprising a reputation-building phase followed by choice of partner for cooperative interactions. By simulating evolutionary dynamics, I investigated the fate of strategies of investing in a cooperative reputation by giving unconditionally to someone who could never reciprocate versus withholding help; of selecting a partner with a good reputation or taking an arbitrary partner; and of cooperating versus defecting in subsequent iterated dyadic interactions.

Results
I found that reputation-building, using reputations in partner choice, and subsequently cooperating could all be favoured. Honesty of reputations was assured when those who invested in a reputation and then defected were subsequently avoided.

Conclusions
I conclude that reputation-based partner choice can be stable when the long term benefits of maintaining honesty outweigh the short term advantages of cheating. I discuss how this differs from costly signalling theory. I also present supporting evidence from economic games in humans which demonstrates how investment in cooperative reputations can reap long term rewards through access to more profitable partnerships. I conclude that reputation-based partner choice provides an important alternative to indirect reciprocity as an explanation for being seen to be cooperative.

Monday, 25th: 10:50-11:15, Aula

Parental investment in child health in Sub-Saharan Africa

Caroline Uggla[1](caroline.uggla.09@ucl.ac.uk), David W. Lawson[1], Ruth Mace[1]
[1]University College London

Objective
Throughout much of sub-Saharan Africa poor child health remains common, yet public health understanding of variation in parental investment in available health technologies is limited. Evolutionary anthropologists on the other hand have generated many hypotheses with regard to factors that influence the optimal allocation of parental investment, but tests of such predictions have largely focused on single community samples and have more often utilised child outcome data (e.g. survival) rather than directly measure parental investment. Here, using data from national household surveys, we examine how characteristics at the level of the child, the mother and the local ecology influence use of curative and preventative health measures.

Methods
We use data from African Demographic and Health Surveys with > 40 000 mothers from 17 countries to construct logistic multilevel models for four behaviours which reduce risk of child death; malaria bed net use, child immunization, prompt treatment of fever and treatment of diarrhoea. We test effects of maternal age, education, wealth and health, child's sex, birth order and health and adjust for overall mortality and the access to health care.

Results
We find that higher maternal age, education and wealth increase odds of investment across all four behaviours. Later born children have a disadvantage in lower odds of immunization and bed net usage and girls have lower odds than boys for fever and diarrhoea treatment. Interestingly, severely malnourished children have higher odds for treatment but lower odds than healthy children for preventative health-seeking behaviours.

Conclusion
We demonstrate consistent patterns of investment in child health based on maternal age, education and wealth across all four outcomes. Other effects vary significantly between health-seeking behaviours. Potential reasons behind these results and the importance of including a wide set of child and mother characteristics will be discussed.

Monday, 25th: 11:15 - 11:40, Aula

What can survey question-ordering experiments tell us about plasticity in human reproductive decision making?

Paul Mathews[1](pmathews@essex.ac.uk), Rebecca Sear[2]
[1]University of Essex, [2] London School of Hygiene and Tropical Medicine

Objective

One of the predictions of evolutionary theory is that selection pressures will endow complex organisms, including humans, with capacities for phenotypic behavioural plasticity. This plasticity allows them to adaptively allocate resources between somatic investment (continued growth and survival), current and future reproduction as well as a trade-off between offspring quantity and offspring quality. Previous research has shown that there are physiological proximate mechanisms for inducing adaptive plasticity in reproduction (e.g. resource scarcity leads to the temporary suppression of ovulation). Here we explore the extent to which psychological mechanisms might be involved in reproductive plasticity. Do subtle primes, in the form of preceding survey questions, change attitudes towards future childbearing?

Methods

We use two experiments which manipulate the ordering of questions in a survey. The first experiment tests the effects on childbearing intentions of preceding questions on mortality and morbidity, using an internet experiment given to students (n=2365) at UK higher education institutions. The second experiment tests for an effect on childbearing intentions of preceding questions on close friends and family using participants (n=409) from the Innovation Panel of the UK Household Longitudinal Study, a broadly representative survey of the UK population.

Results

The first experiment shows that for males, but not females, fertility preferences are significantly higher for individuals primed with questions on mortality and morbidity compared to the control group. The second experiment shows that unmarried participants significantly increase their expectation of having a child after questions on close friends and family compared to controls.

Conclusions

Given the significant results found using two different question-ordering experiments, we conclude that reproductive attitudes are responsive to subtle primes. This suggests that human reproductive plasticity may occur through psychological as well as physiological proximate mechanisms.

Monday, 25th: 11:40-12:05, Aula

Parent-child proximity and women's reproductive fitness in Europe: Testing the predictions of the cooperative breeding and local resource competition models

Susan Schaffnit[1] (susie.schaffnit@lshtm.ac.uk), Rebecca Sear[1]
[1]London School of Hygiene and Tropical Medicine

Objective

The cooperative breeding and local resource competition models make unique predictions regarding the effects of close kin proximity on women's fitness outcomes. The former predicts positive effects of kin proximity on fitness, due to helping between kin; the latter predicts that proximity may reduce fitness if kin are competing over a shared resource base. We test whether co-residence between parents and adult children, and parental survival status influence children's fitness outcomes in high-income countries, and if these effects interact with socioeconomic status. Fitness is measured both by lifetime childlessness and progression to first birth, as timing of births and childlessness are good predictors of overall fitness in low-fertility populations.

Methods

We used Generations and Gender Survey data from nine European countries. First, a multi-level logistic regression was run for women over age 45 with lifetime childlessness as the outcome. Secondly, a multi-level discrete-time event history analysis was run for progression to first birth for all women. Both models used co-residence with parents and survival status of parents to measure kin association. Control variables included those related to women's socioeconomic position, number of siblings, and partnership history.

Results

We found that living with parents at older ages has a consistently negative effect on women's fitness, though the effect is felt more strongly by poor women when considering first birth progressions. Parental survivorship had no effect on childlessness, yet hastened first births.

Conclusions

The positive effects of parental survivorship but negative effects of close proximity on fitness lend some support to both hypotheses.

Monday, 25th: 10:50-11:15, Auditorium

Structure in language is a compromise between compression and expression

Kenny Smith[1] (kenny@ling.ed.ac.uk), Monica Tamariz[2], Hannah Cornish[1], Simon Kirby[1]
[1]University of Edinburgh, [2]University of Granada,

Objectives

Human language exhibits structure: we convey complex meanings using complex, rule-governed utterances. While linguistic structure could be due to biological evolution of the language faculty, languages themselves also evolve on a cultural timescale, as a consequence of their transmission and use. Language learning introduces pressure for compressibility: simpler languages are easier to learn. Language use introduces pressure for expressivity: large, unambiguous languages are more communicatively functional. We use a computational model to identify the conditions under which linguistic structure evolves culturally.

Methods

We model individuals as rational Bayesian learner/users, with a plausible compression-based prior preference for simple languages and an expressivity-driven tendency to avoid ambiguous utterances during language use. We model three population types: diffusion chains (language transmitted to naïve individuals but not used for communication); closed groups (fixed set of individuals communicate repeatedly, no transmission to naïve individuals); replacement microsocieties (language used for communication and transmitted to naïve individuals).

Results

In diffusion chains, pressure for compressibility but not expressivity leads to degenerate languages which are easy to learn but not communicatively functional. In closed groups, pressure for expressivity but not compressibility leads to communicatively-functional but unstructured languages. Only in replacement microsocieties, where both pressures are at play, do we see structure emerge: structured languages are compressible and yet communicatively functional. This model explains a similar pattern of results in experiments involving transmission and use of artificial languages in human populations (Kirby, Cornish & Smith, 2008; Kirby, Tamariz, Cornish & Smith, submitted).

Conclusions

Structure in language, and potentially other culturally-transmitted human behaviour (e.g. music, art), is a result of pressure for learnability and expressivity. The relative importance of expressivity and learnability depends on the structure of populations: the model predicts that differences in the social structure of human populations will yield differences in the structure of human behavior.

Monday, 25th: 11:15-11:40, Auditorium
Selfish hunts: A cultural darwinian analysis of witch persecutions

Steije T. Hofhuis (steije.hofhuis@gmail.com)
Independent researcher

Objective
Although cultural Darwinian theory is gaining popularity, critics claim that the notion of Darwinian selection in cultural evolution is unlikely to provide us with any new scientific insights. However, in this presentation, I argue that cultural Darwinism can indeed resolve sociological puzzles; this is demonstrated by means of a cultural Darwinian analysis of the European witch-hunts.

Methods
Two central and unresolved questions concerning witch-hunts will be addressed. From the 15th to the 17th centuries in Europe, a remarkable concept of witchcraft took shape. The first question is by what or whom was it constructed? This concept contains many elements that appear to be intelligently designed to ensure the continuation of the witch-hunts, such as the witches' sabbath, the diabolical pact, a witch's hidden identity, nightly flight, andtorture as a means of interrogation. The belief in the witches' sabbath, for instance, implied that witches could know each other, which created the possibility of a chain of accusations. The second question is how could witch-hunts reproduce successfully despite the fact that, as many historians have concluded, no one appears to have substantially benefited from the witch-hunts?

Results
Several historians have convincingly demonstrated that witch-hunts were not inspired by an intelligent hidden agenda; it appears that persecutors genuinely believed in the danger of witchcraft. A better explanation of the apparently well-designed concept of witchcraft could be a Darwinian process in which cultural variants were selected and accumulated that accidentally enhanced the reproduction of the witch-hunts. Drawing inspiration from Dawkins's "memetics", witch persecutions also appear to be a socio-cultural phenomenon that reproduced "selfishly". Witch-hunts benefited no one, apart from the witch-hunts themselves.

Conclusions
Cultural Darwinian theory can thus offer us at least two ground-breaking insights. As a process of Darwinian selection, cultural evolution leads to cultural "design without designer" and cultural phenomena can evolve "selfishly".

Monday, 25th: 11:40-12:05, Auditorium
On the reliability of unreliable information: Gossip as cultural memory

Dominic Mitchell[1] (dm362@bath.ac.uk), Gordon P. D. Ingram[2], Joanna J. Bryson[1]
[1]University of Bath, [2]Bath Spa University

Objective
Humans continue to trust socially-learned information even when it is unreliable and even in the face of direct evidence of the senses to the contrary. Our objective is to provide a model which can explain such seemingly maladaptive behaviour in evolutionary terms.

Methods
We focus on one class of such information: gossip. We use an Agent Based Model where a population of agents play the donation game. Cooperation can evolve by indirect reciprocity when agents observe others' behaviour in the game, evaluate it according to a social norm, spread the information by gossip and act on it. We first introduce error in the observation of gossip, and then in the transmission (gossiping). We compare two strategies: relying on observation, or on gossip. We introduce a limited lifespan for agents so that information gleaned might never be put to use.

Results
We confirm earlier theoretical work that cooperation can still evolve when the evaluation of others is sourced from observation containing a high degree of uncertainty. The boundary is expressed by the equation $1/q = b/c$. where q is uncertainty and b,c benefit and cost respectively in the donation game. Our second result is that when transmission error is high gossip can still enable cooperation provided the rate of transmission is fast enough. Our third result is that in a situation where high transmission error in gossip renders direct observation the better strategy, if agent lifespan is reduced the outcome is reversed and relying on gossip becomes the winning strategy despite its high rate of error.

Conclusions
Reputation by gossip is high on relevance but low on veracity, whereas for sense evidence the reverse is true. But these scores are sensitive to the group's ecology. The spread of and reliance on error-prone gossip may well be adaptive.

Monday, 25th: 13:30-13:55, Auditorium
On the Excludability of Public Goods

Daniel Taylor[1] (djt20@bath.ac.uk), Marios Richards[2], Joanna Bryson[1]
[1]University of Bath, [2]Independent Scholar

Objective
It is often stated that reciprocation is insufficient to explain the evolution of cooperation in human populations. We argue that this position is based, in part, on a set of unnecessarily harsh modeling assumptions - namely, that humans regularly engaged in activities to the non-excludable benefit of the rest of the group. We demonstrate that if these assumptions are relaxed, then reciprocation can explain the evolution of two of the most highly cited example of cooperative activities which humans engaged in pre-(?) history - food sharing and warfare.

Methods
We develop a simple model of cooperation where the benefits of cooperation are excludable. Excludability is an exogenous variable which measures the degree to which the benefits of a public good can be excluded from a given individual. We then compare these modeling assumptions to empirical studies of food sharing amongst the Ache of Paraguay, and warfare among the Turkana of Kenya.

Results
For the Ache, we find that food is highly excludable, and that reciprocation is sufficient to explain cooperation. With the Turkana, we find that the benefits of war – loot (cows) – means that this is not a prisoner's dilemma at all, but is simply a beneficial activity in which to engage.

Conclusion
We conclude that there is no evidence that either of the canonical examples of cooperation conform to the modeling conditions required to undermine reciprocation as an explanation of human cooperation. Considering the volume of theoretical work aimed at explaining cooperation, we argue that it is vital to establish that such a behaviour actually exists

Monday, 25th: 13:55-14:20, Auditorium
Facial appearance influences offers made in the Ultimatum Game

Poppy I. Mulvaney[1] (P.mulvaney@bristol.ac.uk), Ian S. Penton-Voak[1]
[1]University of Bristol

Objective
We investigated how the facial appearance of the receiver influences offers made in the ultimatum game. Although considerable work has investigated individual differences in proposer behaviour in this game, little work has investigated the effects of receiver traits on proposer's offers.

Methods
In study 1, 42 participants were asked to play as the proposer in a series of one-shot ultimatum games; the proposers saw facial images of the male virtual receivers in which masculinity had been manipulated. In study 2, UK participants (N=72) played a similar ultimatum game against a series of virtual opponents represented by unmanipulated male facial images; they also assessed each image for 'trustworthiness' and 'formidability' – two dimensions that capture much of the variance in trait attribution to faces. Study 3 was a replication of study 2 (N=157), run in an American university.

Results
In study 1, UK proposers made more fair offers to masculine than feminine faced receivers. In study 2, rated 'formidability' was a strong predictor of fair offers made; rated 'trustworthiness' had no effect on offers made. In study 3, in contrast to our UK undergraduate population, the American data showed that 'trustworthiness' was significantly associated with fair offers, and was a stronger predictor of offers made than 'formidability'.

Conclusions
For all studies we demonstrate that facial appearance of receivers plays an important role in proposer behaviour in the Ultimatum Game. We also show that although the trait attributions made to faces were statistically indistinguishable between the UK and the US in studies 2 and 3, the role of these attributions in economic decision making varies cross-culturally, even between WEIRD cultures.

Monday, 25th: 14:20-14:45, Auditorium
System-wide structure emerges through cultural transmission

Hannah Cornish[1](hannah@ling.ed.ac.uk), Simon Kirby[1], Kenny Smith[1]
[1]University of Edinburgh

Objectives
Language is a culturally transmitted behaviour which displays systematicity: linguistic rules interact to create cohesive and often complicated organisational patterns reflective of system-wide structure. Theories differ as to why this should be the case, with some researchers emphasising the role of functional (external) pressures on language, and others emphasising language-specific (internal) pressures. We investigate the possibility that system-wide structure can emerge as a result of very general, non-linguistic constraints on sequence learning, in conjunction with a process of cultural transmission

Methods
Learners were asked to reproduce, one at a time, a set of 60 sequences of flashing colours (red, blue, yellow and green). Crucially, on each presentation, participants were shown a single sequence and asked to reproduce it immediately. Participants worked through the set of 60 sequences twice, with the sequences they entered on the second pass through the set being transmitted to a subsequent learner in a diffusion chain (the initial participant was exposed to a randomly constructed set of length twelve sequences). We ran 4 such chains, each of 10 participants.

Results
Across all chains, sequences evolve gradually over generations to become easier to learn. In addition, sequences within a given chain became more self-similar, indicating the emergence of lineage-specific systematic patterns. Analysing these patterns revealed the presence of hierarchical structure, despite there being no such structure initially, and despite the task involving only immediate recall of individual, independent sequences.

Conclusions
We show that system-wide structural features naturally emerge from a non-linguistic sequence-learning task when that task is iterated. This is the first demonstration in an implicit learning task of the cumulative cultural evolution of a set of initially independent behaviours becoming systematic.

Monday, 25th: 14:45-15:10, Auditorium
Female economic dependence and the morality of promiscuity

Michael E. Price[1](michael.price@brunel.ac.uk), Nicholas Pound[1], Isabel Scott[1]
[1]Brunel University

Objective
Sexual morality varies widely across individuals and social groups, and differences in sexual morality underlie important cultural and political conflicts in many societies (e.g.,the USA). We attempt to account for variance in a particular kind of sexual morality—anti-promiscuity sentiment—with a model based on the evolutionary concept of paternity certainty. According to this model, people tend to express more anti-promiscuity sentiment in environments in which females are more dependent on male parental investment. This relationship should exist because when male parental investment is more essential, promiscuity—which undermines paternity certainty—should entail greater costs and reduced benefits for both sexes.

Methods
We tested the paternity certainty model in two survey studies involving US participants recruited via MTurk. In both study one (N = 656) and study two (N = 4,533), we measured participants' anti-promiscuity sentiment and their perception of how much females in their environment depend economically on a male mate, as well as control variables including religiosity and political conservatism. In study two we collected sufficient data from all 50 US states to evaluate how well economic indicators of female economic dependence (e.g. female income, availability of welfare benefits) predict anti-promiscuity sentiment at the state level.

Results
The paternity certainty model of anti-promiscuity sentiment was supported in both studies. At the individual level of analysis, female economic dependence on a male mate was a significant positive predictor of anti-promiscuity sentiment, even after controlling for the effects of religiosity and conservatism on this sentiment. At the state level, female income was a significant negative predictor of anti-promiscuity sentiment, and this relationship was fully mediated by female economic dependence on a male mate.

Conclusions
The paternity certainty concept seems to provide a useful framework for predicting variance in anti-promiscuity sentiment, across both individuals and social groups.

Monday, 25th: 15:50-16:15, Auditorium

Increased mortality exposure within the family rather than individual mortality experiences trigger faster life-history strategies.

Charlotte Störmer[1](charlotte.stoermer@phil.uni-giessen.de), Virpi Lummaa[2]
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Objective

Life History Theory predicts that extrinsic mortality risk is one of the most important factors shaping (human) life histories. While evidence from contemporary populations suggests that individuals confronted with high mortality environments show characteristic traits of fast life-history strategies, little is known of the impact of mortality experiences on the speed of life histories in historical human populations with generally higher mortality risk, and on male life histories in particular. Furthermore,it remains unknown whether family or individual level mortality experiences have a greater effect on life-history decisions in humans.

Methods

In a comparative approach using event history analyses, we study the impact of family versus individual level mortality exposure on two central life-history parameters, ages at first marriage and first birth, in three historical human populations (Germany, Finland, Canada). Mortality experience is measured as the confrontation with sibling deaths within the natal family up to an individual's age of 15.

Results

Results show that individuals adjust the speed of life histories according to the mortality experiences of their natal family but not according to individual experiences. The general finding of lower ages at marriage/reproduction after exposure to higher mortality during childhood holds for both females and males.

Conclusion

This study shows that both sexes can - and do -adjust the speed of their life histories in relation to early mortality exposure. It also provides evidence for the importance of the family environment for reproductive timing while individual level experiences seem to play only a minor role in reproductive life history decisions in humans.

Monday, 25th: 16:15-16:40, Auditrorium
Competition for partners drives cooperation among strangers

Arno Riedl[1] (a.riedl@maastrichtuniversity.nl), Aljaz Ule[2]
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Objective
This paper investigates whether and when cooperation can be achieved through selection of partners.

Methods
Groups of nine human subjects play 60 periods of a laboratory prisoner's dilemma game (PD) with varying freedom to select a game partner. In each period subjects are randomly and anonymously matched into three triplets and observe each other's last PD action. In each triplet at most one pair can play the PD. We compare three experimental conditions that vary how this pair is determined. In the BASELINE treatment the pair is chosen randomly. In the REFUSAL treatment the pair is chosen randomly but each chosen subject can then refuse to play the PD. In the COMPETITION treatment each subject first indicates which other subject in her triplet she would refuse and then a feasible pair, if one exists, is randomly chosen. The REFUSAL treatment adds the possibility of exclusion to the BASELINE treatment. The COMPETITION treatment adds the possibility to discriminate between possible partners, which introduces competition for partners. Non-playing subjects earn less than any possibly PD payoff, which makes exclusion and discrimination costly. We observe 10 independent groups of nine subjects in each treatment.

Results
As expected cooperation rapidly declines towards zero in both BASELINE. Unexpectedly this is also the case in REFUSAL, showing that exclusion of unwanted partners by itself does not promote cooperation. In stark contrast, cooperation increases in COMPETITION, confirming that competition for partners can be a cooperation-promoting mechanism among anonymous strangers. However, also in COMPETITION cooperations success is not guaranteed. Whereas some groups (6 out of 10) achieve 100% cooperation in other groups (4 out of 10) it declines towards 0% cooperation. This suggests the existence of two behavioral equilibria under partner competition.

Conclusions
Our results confirm that competition for partners is a feasible alternative to punishment as a mechanism for sustaining cooperation.

Monday, 25th: 16:40-17:05, Auditorium
Are social personality traits calibrated to physical differences in the body and face?

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Objective
It has been suggested that the ubiquitous genetic variation in personality traits like extraversion and shyness is maintained in populations by balancing selection or soft selection on sweeping alleles. However, recent molecular genetic evidence from genome-wide association studies and complex trait analyses might be problematic for such explanations. At the same time, there has recently been renewed interest in an old hypothesis by Tooby and Cosmides (1990) that personality traits are calibrated to own, potentially fitness-related physical characteristics like attractiveness and strength, and that thus personality heritability might be an indirect reflection of the heritability of these characteristics.

Methods
Detailed body and face measurements of 115 male and 120 female young adults were taken using 3D white-light scanners and lab devices. These were used to calculate various indices, including body and face masculinity-femininity, muscularity, upper body strength, fluctuating asymmetry, height, body mass/volume, and aerobic fitness. In addition, attractiveness was assessed with self-reports and observer ratings based on rotating 3D body models. These measures were related to self-reports of social personality traits like extraversion, agentic and antagonistic narcissism, shyness, dominance, and aggressiveness.

Results
For men the overall pattern of results was in the expected direction and there were several interesting associations of anthropometric measures related to attractiveness and strength with social personality traits. However, effect sizes were generally small. There were no systematic findings for women.

Conclusion
While there seems to be some relationships of physical shape and formidability with personality in men (but not in women), the strength of these associations might have been overstated recently. It appears unlikely that heritable physical differences are a major determinant of the genetic variation in personality traits, especially since causal influences might be bidirectional. Implications for the evolutionary understanding of social personality traits are discussed.

Monday, 25th: 17:05-17:30, Auditorium

Morphological masculinity predicts perception of dominance but not health in 3D men's faces

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Objective
The immunocompetence handicap hypothesis is interpreted to imply that masculinity in men's faces acts as cue to health and thus affects attractiveness to women. Masculinity may also have evolved as a cue to dominance in intrasexual competition, hence being indirectly attractive to women as a cue to power and status. To test these alternative accounts we compared morphological measures of facial masculinity to perceptual judgments, determining facial morphology in a way that parallels image manipulations previously employed to study the perceptual impact of masculinity.

Methods
Women judged the masculinity, dominance, and health of 40 colour- and texture-standardised, rotating 3D men's faces (age 18-26, BMI 18-25) in three separate studies. We calculated morphological masculinity using: (a) previously employed discriminant scores and (b) "average" scores based on the shape vector defined by the average difference between men and women's facial shapes. Average scores were computed similarly for facial morphology associated with body weight and height since these attributes may contribute to masculinity perception.

Results
(1) Morphological masculinity (as measured by "average" but not discriminant scores) predicted perceived masculinity and dominance but not health. (2) The morphological correlate of weight was an independent and stronger predictor of perceived masculinity and dominance than morphological masculinity, while perceived masculinity was also affected by height

Conclusion
Our findings show that a morphological measure of faces can be a useful index of perceived masculinity. They also show that effects of height and weight need to be considered when assessing the perceptual impact of sexual dimorphism. That a morphological measure of sexual dimorphism was associated with the perception of dominance but not health, suggests masculinity in men's appearance plays a more important role in intrasexual competition than in intersexual selection of health cues.

Tuesday, 26th: 09:30-09:55, Aula

Copy when uncertain: Neuronal correlates underlying the use of social information.

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Objective

Social information exerts a strong influence on decision-making through an integration of individual experiences with information derived from the actions of others. It has been proposed that the extent of social information use depends on one's own uncertainty according to a copy-when-uncertain strategy where information from individual and social sources is combined based on their respective reliabilities.

Methods

We investigate this integration process by extending models on Bayes optimal integration of sensory information in a social decision context. We then use a model based fMRI appraoch to identify the neural substrate linked to the fidelity of this integration process.

Results

We show that a majority of individuals manifest near Bayes optimal behavior when integrating two distinct sources of social information sources but systematically deviate from Bayes optimal choice when integrating individual and social information. This systematic behavioural deviation was linked to activity in the left inferior frontal gyrus where individuals who underuse social information show less activity.

Conclusions

Thus, exploiting social information requires additional cognitive resources that allow for executive control over decision-making processes that are initially biased towards individual information. The recruitment of these top down processes is modulated by the uncertainty of individual information. The results suggest that uncertainty strongly influences social information use and thus provide a mechanistic explanation for the hitherto puzzling findings where human individuals neglect social information.

Tuesday, 26th: 09:55-10:20, Aula

Evolution of prospect theory: An adaptive account of variations in our willingness to take risks

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Objective

How humans choose between risky options is described by prospect theory (Kahneman & Tversky, 1979), which has been hugely influential in the social sciences. One common finding is that individuals are risk averse for highly probable gains and unlikely losses, but risk seeking for highly probable losses and unlikely gains. This fourfold pattern of risk attitudes is a well-established empirical finding but it currently lacks a functional, evolutionary explanation.

Methods

We explored individual variation in risk preferences by modelling an individual in a changing environment in which it has to acquire energy to prevent starvation and build up enough reserves to reproduce. We compute future reproductive value through dynamic programming, and by offering a one-off choice between a stochastic and deterministic option of equal expected value, we can identify risk preference for an individual who is behaving optimally.

Results

Under reasonable conditions, our model successfully predicts the commonly observed fourfold pattern of risk preferences: not only risk aversion for gains and risk seeking for losses, but the ‘lottery effect’ and ‘insurance effect’ associated with unlikely outcomes. This pattern critically depends on some level of background mortality (death from disease and predation), which creates a level of urgency that promotes appropriate risky behaviour even when times are good.

Conclusions

Our model provides a valid evolutionary account of prospect theory, based on optimal foraging and risk sensitivity. This suggests that the way that people make decisions when faced with risky options may represent a rational, adaptive response to a stochastically changing world.

Tuesday, 26th: 10:50-11:15, Aula

Lost letter measure of variation in altruism and parochialism in 30 neighbourhoods

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Objective

Human cooperative behaviour varies across human populations and this variation is likely to be explained by variation in both the ecological and cultural context of the populations. Most studies have focused on measuring variation between cultural groups using experimental economic games and there is a lack of sampling of within group variation based on naturalistic measures of cooperation. In this study we determine which area level socio-economic and cultural characteristics predict the variation in altruism and parochialism of individuals living in an area.

Methods

We measured variation in altruism and parochialism using the lost letter technique. We dropped letters on the pavement addressed to fictitious neutral and sectarian charities in 30 different Catholic and Protestant neighbourhoods with a wide range of religious composition and socio-economic characteristics in Belfast, UK.

Results

The results show a negative effect of neighbourhood income deprivation on altruism with letters dropped in the poorest neighbourhoods being less likely to be returned than letters dropped in the wealthiest neighbourhoods. We also find increased levels of parochialism in deprived neighbourhoods, although this effect is only present in predominantly Protestant neighbourhoods with the difference in the number of Protestant and Catholic letters returned being significantly higher in deprived Protestant neighbourhoods.

Conclusions

We suggest that measures of cooperative behaviour are strongly context dependent with both socio-economic and cultural factors likely to play an important role in determining variation in altruism and parochialism.

Tuesday, 26th: 11:15-11:40, Aula

Social learners require process information to outperform individual learners

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Objective

Humans exhibit a rich and complex material culture with no equivalent in animals. Social learning, a crucial requirement for culture, provides a means to accumulate knowledge over time and to develop advanced technologies. Despite the growing interest in cultural evolution, little attention has been focused on the relation between social learning and the complexity of human material culture. Here, the aim was to compare the efficiency of individual learning and two types of social learning (product copying and process copying) faced with a complex and opaque virtual task.

Methods

120 participants, placed in closed groups of 5 peoples, played a computer game during which they had to achieve a complex virtual task. The aim was to build a virtual fishing net to capture fish during virtual fishing trials. Each of the 15 successive trials was followed by an information period. The level of social information shared between the players differed according to the treatments. For the individual learning treatment, the players had access only to the last trial and the cumulative scores of each of their other group members. For the product-copying treatment, the players could additionally see the nets, associated with their scores, of their group members. Finally, in the process-copying treatment, the participants could see the step-by-step process associated with the net.

Results

Individuals from process copying groups outperformed individuals from product copying groups and individual learners, whereas access to product information was not a sufficient condition for providing an advantage to social learners compared to individual learners.

Conclusions

The importance of process-copying for efficient information transmission between individuals has important implications for understanding cumulative cultural evolution. Indeed, it was argued that most human artefacts are more or less opaque, even in relatively simple material cultures such as those of hunter gatherers, suggesting that cumulative cultural evolution could require process-copying ability.

Tuesday, 26th: 11:40-12:05, Aula

Consistent individual differences in human social learning strategies

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Objective

Social learning has been key to human evolutionary success. The cultural transmission of information between individuals by social learning enabled human groups to rapidly adapt to a broad range of habitats across the globe. Our capability of learning from our peers allowed us to build on existent knowledge and effectively employ numerous forms of group cooperation and coordination. However, not much is known about the underlying mechanics of human social learning in contexts where payoffs of behaviour depend on the behaviour of others. Here we experimentally study social learning strategies across contexts of human interaction.

Methods

We conducted an economic experiment in which people made decisions in groups of eight. Subjects made binary decisions in four repeated games: i) a game where one option yielded higher expected payoffs, ii) a social dilemma, iii) a coordination game and iv) an evasion game. Payoffs were noisy, making it harder to find out which option was better. Before making their decisions, subjects could collect costly information about the decisions and associated payoffs of their peers.

Results

We show that humans consistently differ in their social learning strategies across interaction contexts. Many individuals predominantly base their decisions on behavioural information only, irrespective of the interaction context, whereas others consistently rely on information on the payoffs associated with behaviour. We assess the consequences of this variation for cultural evolution with a simple model, showing that groups harbouring individual differences in social learning strategies can adopt superior technologies more readily, and can maintain behavioural diversity in an evasion game.

Conclusions

Our results call for new theory shedding light on how variation in social learning strategies may evolve by natural selection. In addition, this study highlights the need for exploration of the consequences of personality differences in social learning strategies for the evolution of cultural traits.

Tuesday, 26th: 10:50–11:15, Auditorium

Health care hypothesis better predicts variation in percentages of left-handers than the fighting hypothesis.

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Objectives

In human societies left-handers are always in the minority relative to right-handers and associated with fitness costs. What then explains its persistence in human populations? The fighting hypothesis proposes that left-handers have an advantage during fights and are subject to negative frequency dependent selection. This is supported by (1): an over-representation of left-handers in interactive sports; (2) a comparative study showing that among non-industrial societies the proportion of left-handers positively correlates with the incidence of homicide (Faurie and Raymond 2005). We provide new data for both types of evidence and suggest another potential explanations for the persistence of left-handedness.

Methods

To test the over-representation of left-handers in different types of sports we performed a meta-analysis over 134 sports. For the second line of evidence we measured handedness in the population with one of the highest recorded left-handedness (Eipo, Papua-Indonesia) as measured from indirect evidence by Faurie and Raymond (2005). To test an alternative explanation we analysed among 12 Western populations the relationship between left-handedness and public expenditure on health care and homicide respectively.

Results

The meta analysis supported the over-representation of left-handers in specifically combat sports, but no (indirect) evidence for the postulated increase in winning changes of left-handers. In the non-industrial society we found extreme low levels of left-handedness, contrary to the earlier estimate. Among western societies, the proportion of left-handers positively correlated with public expenditure on health care, but not with frequency of homicide.

Conclusions

The evidence for the fighting hypothesis is currently weak. The persistence of left-handedness may be partly the by-product of pathology on which selection pressures cannot act upon. Reasons for overrepresentation of left-handers in combat sports are discussed. Based on fitness studies we suggest that not only the evolution of direction but also of strength of handedness should be considered.

Tuesday, 26th: 11:15-11:40, Auditorium

Impact of attachment on adults and childrens face preferences

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Objectives

Cognitive models of face recognition suggest that the structural information of the observed face is delivered on two pathways; one leading to the brain regions of person identification, and a second leading to the area of affective responses. Detecting familiar faces may involve both of these mechanisms. Sexual imprinting model predicts that ones spouse resembles ones opposite-sex parent when emotional commitment was high during childhood. We hypothesize that positive emotions coming from a favoring family environment mediate cognitive processes of face preferences. Our aim was to gather behavioral evidence for emotions influencing adult face preferences, and to investigate whether this influence is already present in children.

Methods

From individual photos of adults and preschool children composite faces were constructed, and transformed in shape, in order to resemble either subjects parents or an unknown individual. Adults (N=96) were exposed to opposite-sex image pairs, and were instructed to choose the most attractive one. Children (N=87) between 3 to 6 years were asked to show which one of the same-sex transforms they found more appealing: the familial- or the control face. Attachment was measured by the EMBU retrospective questionnaire, and a projective test, respectively.

Results

Adults with good childhood experiences preferred faces resembling their opposite-sex parent more than those with more problematic relationship. Childrens preference was also mediated by the attachment style: children with weaker attachment to their parents were more likely to choose the control face than the father-resembling one.

Conclusions

Emotional proximity to a familiar individual influences preference for unknown others who resemble this person. This bias may affect social decisions during the whole lifetime. These results provide initial support for a proposed model explaining the neural background of the preference for familiarity.

Tuesday, 26th: 11:40-12:05, Auditorium

The behavioral immune system and anti-immigration attitudes: Individual differences related to disgust shape opposition to immigration

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Objective

Why does the political issue of immigration ignite so strong sentiments? We argue that this is so because immigrants activate biological, visceral mechanisms designed for disease avoidance that operate through disgust and are designed to remain active even in the face of the kinds of good intentions and signals of honesty that would otherwise form a basis for peaceful co-existence.
Recent research suggests that the immune system has a behavioral component that motivates us to avoid potential infectors and that disgust is a critical motivational component of this system. Over evolutionary history, a key source of pathogens has been other people and a key feature of the adaptive cognitive mechanisms designed to gauge infection risk in others is to treat any non-prototypical physical appearance as a cue of potential pathogen infection. Given this it is likely that immigrants – in particular from ethnically different countries – would activate the behavioral immune system. Importantly, this system should be highly sensitive to all information suggesting that an individual has a background in a different pathogen ecology. This would relate to birth place and other factors indicative of the deep background of an individual. Importantly, these inferences should be insensitive to information that other social motivational system are highly sensitive to: information about respect for democracy and intentions to contribute to the community through hard work.

Methods

We test this argument in a nationally representative survey experiment and in nationally representative cross-sectional data collected in the US and Denmark and involving over 1000 individuals.

Results and conclusions

We demonstrate that disgust-sensitive individuals hold stronger anti-immigration attitudes and assign greater importance to factors indicative of the deep background of an individual when making in group/out group classifications. Importantly we also demonstrate that these effects of disgust endure even facing cues that have proven to dampen opposition towards immigration in prior research but are unassociated with the behavioral immune system.

Tuesday, 26th: 14:30-14:55, Auditorium

Social stratification and hypergyny: Towards an understanding of male homosexual preference

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Objective

Male homosexual preference (MHP) challenges evolutionary thinking because the preference for male-male relationships is heritable, implies a fertility cost (lower offspring number), and is relatively frequent in some societies (2-6% in Western countries). Proximate explanations include the hypothesis of a "sexually antagonistic factor" in which a trait that increases fertility in females also promotes the emergence of MHP. Because no animal species is known to display consistent MHP in the wild (only transient and contextual homosexual behavior has been described), additional human-specific features must contribute to the maintenance of MHP in human populations. Social stratification is a human specific feature that appeared recently in some societies. It has important implications since it affects the pattern of marriages, leading to class endogamy and hypergyny. The aim was to test if social stratification together with hypergyny could have promoted the emergence and maintenance of MHP in Humans.

Methods

A theoretical model was built to study the effect of social stratification and hypergyny on the evolution of a sexually antagonistic factor inducing MHP. Computer simulations were then performed with various numbers of classes, conditions of demographic regulation, and mating systems. Finally, anthropological data were collected to test the link between the probablity to find MHP in a society and the level of social stratification.

Results and Conclusions

In a stratified society, a relatively high frequency of MHP could be maintained as a result of the social ascension of females signaling high fertility (hypergyny). The prediction that MHP is more prevalent in stratified societies was supported. More generally, any traits associated with up-migration are likely to be selected for in stratified societies and will be maintained by frequency-dependence even if they induce a pleiotropic cost. These results offer a new perspective for understanding seemingly paradoxical traits in human populations.

Tuesday, 26th: 14:55-15:20, Auditorium

The co-evolution of social institutions and large-scale human societies

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Objective

Human social interactions are typically regulated by institutions. These institutions effectively create the rules of the game that individuals play, by determining the payoff for different social actions. The evolutionary origin of such institutions has, however, remained largely unaddressed. We address here the role that institution formation can play in the transition from small- to large-scale human societies.

Methods

We model a structured population of individuals that live on discrete patches. Each patch can contain both social individuals that create a sanctioning institution, and asocial individuals that remain outside of an institution. Individuals carry two cultural traits. The first determines whether they either a) remain asocial, b)join a social group on their patch and contribute to a public good (cooperate), or c) join a social group but do not contribute (defect). The second trait is a preference for the type of institution. Specifically, for the proportion of the group's public good that should be invested in sanctioning non-contributors. A group's institution is then constructed from an aggregation of the group members' preferences. The remainder of the public good not assigned to sanctioning is invested in increasing the carrying capacity of the group, e.g. through technological improvement.

Results

Social individuals are able to invade a population of asocials, and establish sanctioning institutions that select for high levels of cooperation, provided that patch size is initially small. The benefits of cooperation then raise carrying capacity and hence drive a large increase in group size. Crucially, once the institution is established, it is able to maintain cooperation even as group size subsequently becomes very large.

Conclusions

Cooperation promoting institutions can evolve in small groups, and can then maintain cooperation as group size expands. The coupling of institution creation with demography can thus help to explain the transition from small- to large-scale societies.

Tuesday, 26th: 15:50-16:15, Auditorium

A not-so-grim tale: How childhood family structure influences reproductive and risk-taking outcomes in a historical population

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Objective

Family structure during childhood plays an important role in shaping a child's life course. Most empirical studies have examined the influence of one or other type of family disruption or composition. This makes it difficult to simultaneously assess the influence of different family structures on later child outcomes.

Methods

We use a large, rich data source collected by Alfred Kinsey and colleagues in the United States from 1938 to 1963 to examine in detail the effects of particular childhood family compositions and compare between them (n=16207). It further allows us to look at an array of traits relating to sexual maturity, reproduction and risk-taking behaviour.

Results

Our results show that for women, the absence of either natural parent from the childhood home accelerates reproductive events such as age at first sex, first marriage and first birth, although the effects are stronger for father absence than for mother absence. Further, the addition of a stepfather to the household appears to play a key role, as sexual behaviour was accelerated once a stepfather was introduced, compared with living with a single mother only. Also among men, mother -but not father- absence is associated with later age at puberty. These results withstand adjustment for socio-economic status, age, ethnicity, age at puberty (where applicable) and number of siblings.

Conclusions

While these results are in line with the hypothesis that early family environment influences later reproductive strategy, the different responses to the presence or absence of different parental figures in the household, and the somewhat different responses of the two sexes, suggests that parental absence and alloparents represent more than simply an unstable rearing environment but that particular family constructions exert independent influences.

Tuesday, 26th: 16:15-16:40, Auditorium

The evolution of cooperation in harsh environments, with emphasis on cooperative breeding behavior in early hominins

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Objective

Since Kropotkin's Mutual Aid, researchers have posited that cooperation should increase in harsh environments, but this claim still lacks theoretical underpinnings. Understanding how increased cooperation evolves in environments of increasing adversity is an important puzzle, particularly since the recent suggestion that the adoption of cooperative breeding behavior by early hominins was a prerequisite stage in the evolution of our large brains.

Methods

First, my collaborators and I analyzed a game-theoretic agent-based model in which benefitting from altruistic behavior was essential for survival and reproduction. This highlights a conflict between the short- and long-term rewards of cooperation. In the long-run, cooperation is favored because only groups with sufficient numbers of cooperators will survive. In the short-run, however, environmental costs increase the advantage to defectors in cooperator-defector interactions. Next, we developed a multi-tiered model, based on details of human natural history and the social structure of small-scale societies, to investigate the evolution of cooperative breeding behaviors. Agents' lives were segmented into key life stages and agents were organized into family groups that could cooperate on childrearing. The influences of three forms of environmental harshness were assessed: (1) increased costs of childrearing, (2) decreased ability of individuals to genetically adapt, and (3) temporal variability in the costs of childrearing.

Results

Harsher environments can select for increased cooperation in the long run, as long as a mechanism for promoting group assortment is in place. We find support for the proposal that harsh environments select for cooperation in childrearing. Cooperator frequencies increased under higher costs of childrearing as well as under increased restrictions to genetic adaptability.

Conclusions

Our results shed new light on the evolution of cooperation via interdependence and suggest that the ability of humans to survive in an environment of increasing scarcity is linked to increasing cooperation in the raising of offspring.

Tuesday, 26th: 16:40-17:05, Auditorium

Is phenotypic development guided by forecasts? A test of adaptive developmental plasticity in pre-industrial Finland

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Objectives
It is known that environmental experiences during early life have profound consequences for human health and psychology in adulthood. This is often interpreted as evidence of adaptive developmental plasticity, whereby development of the phenotype is adaptively guided by a ‘weather forecast’ of external conditions. Such explanations are widespread in academic fields seeking to explain human variation, from evolutionary psychology to the biomedical sciences. This kind of adaptive developmental plasticity has been well documented in a variety of wild and laboratory animal populations, but it is unclear to what extent it might be true in long-lived species such as humans in which the long time delay between development and adulthood may decrease the likelihood that forecasts will be accurate. One key prediction of this general hypothesis, which has not been empirically tested, is that evolutionary fitness should be higher when individuals experience an adulthood environment that matches that which they experienced during early life than when they experience an adult environment that does not reflect their early experience.

Methods
We used detailed longitudinal demographic and food availability data from a historical Finnish population with large variation in annual crops yields and subsequent mortality to test whether survival and reproductive success of individuals during severe famine depended on their food availability around birth.

Results
We found that, contrary to predictions derived from hypotheses of developmental plasticity, individuals experiencing low food availability during development showed both lower survival and reproductive success during the famine than those experiencing favorable conditions. These effects were more pronounced among young individuals and those of low socio-economic status.

Conclusions
The results suggest that poor environmental conditions around birth increases the vulnerability of individuals to later adversity and lend support to the argument that hypotheses of adaptive developmental plasticity need to take account of such constraining developmental effects.

Tuesday, 26th: 17:05-17:30, Auditorium

Why are stepfathers bad for child development? Testing the effects of stepfather presence and investment on multiple child outcomes in the UK.

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Objective

Stepfather presence has been found to have detrimental effects on multiple child outcomes in developed countries. At present, the processes behind these effects are unclear, with most studies focusing on the "presence" of stepfathers without taking direct investment levels into account. Therefore, we investigate whether the stepfather effects are due to: 1) lower levels of direct investment by stepfathers compared to fathers; 2) lower levels of direct maternal investment associated with stepfather presence; 3) inferior quality of direct investment by stepfathers compared to fathers; and if 4) stepfather presence itself is a stressor for children.

Method

Using data from the Avon Longitudinal Study of Parents and Children, we carry out regression models to investigate the effects of stepfather/father presence and direct investment on children's height, educational attainment and behavioural difficulties at age 7. Missing values are imputed using "mi impute chained" in STATA SE 12.

Results

Unlike previous studies, we did not find stepfather effects on Height. However, we found stepfather effects on educational attainment and behavioural difficulties. First, we found that the negative stepfather effect on educational attainment is likely to do with lower levels of direct investment provided by stepfathers compared to fathers. Second, the negative stepfather effect on behavioural difficulties is likely due to multiple factors: stepfather presence itself is a stressor, stepfathers invest less, and direct investment by stepfathers is of inferior quality compared to fathers.

Conclusions

Our results show that the negative effects of "stepfather presence" can be due to different processes depending on the child outcomes. Our results imply that the negative stepfather effect on educational attainment may be overcome if stepfathers are encouraged to invest more in their stepchildren. However, the negative stepfather effect on children's behaviour will be difficult to overcome even if stepfathers invest more.

Wednesday, 27th: 09:30-09:55, Aula

Prosocial preferences do not explain human cooperation in public-goods games

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Objective

It has become an accepted paradigm that humans have prosocial preferences that lead to higher levels of cooperation than expected. Our objective was to show that the results of such economic games have been over-interpreted and the pro-social explanations not sufficiently tested.

Methods

We do this by experimentally varying the information that individuals receive about the consequences of their behavior for others. We used a standard public-goods game as a control but also varied the information participants received in additional treatments. In our novel method, we use a treatment the removes all social aspects of the game. Our participants were unaware they were playing a social dilemma, and were presented with a virtual black box that they could choose to invest in. Otherwise the game was identical to the standard game with the players and their payoffs linked as normal. In our enhanced treatment, the players received more information than normal after each round of play - specifically they learned the payoffs of each groupmate. We also repeated all three versions but under conditions where full cooperation is also the best strategy.

Results

(i) the level of cooperation did not differ between the control and our black-box treatment, in which players did not even know their choices affected others; (ii) providing players with enhanced information about the earnings of their groupmates led to lower levels of cooperation, even when cooperation was directly profitable. (iii) when we made 100% cooperation the income maximizing and prosocial strategy, we foundthat cooperation failed to reach 100% in any treatment.

Conclusions

Overall, these results contradict the suggested role of prosocial preferences and show how the complexity of human behavior can lead to misleading conclusions from laboratory experiments. Prosocial preferences are neither necessary nor sufficient to explain the results.

Wednesday, 27th: 09:55-10:20, Aula

Do individual differences in developmental plasticity result from parents hedging their bets? A mathematical model

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Children vary in the extent to which their development is affected by particular experiences (e.g., maltreatment, punishment, social support). Empirical work shows such variation can result from differences in genes and/or differences in prior experiences. From a functional viewpoint, differential susceptibility to environmental influences is puzzling: Is there no single optimal level of plasticity that all children should develop? One influential hypothesis states that different levels of plasticity may be optimal in different environmental states, and because environments are variable, parents do not know what environmental states their offspring will experience (i.e., what level of plasticity will be best for their offspring); therefore, natural selection might favor them producing offspring with varying levels of plasticity. This talk (1) presents a mathematical model examining the logic of this hypothesis—whether conditions exist in which natural selection favors parents to hedge their reproductive bets by producing diverse offspring; and, (2) whether these conditions plausibly apply to humans. Consistent with theory from biology, results indicate that variation in plasticity may be favored if environmental change occurs temporally (not spatially) at particular frequencies, and when fitness effects (the impact of an individual’s plasticity on her total fitness) are large. We argue that parental bet-hedging will be more likely instantiated via epigenetic mechanisms (e.g., pre- or postnatal programming) than genetic ones (e.g., mating with genetically diverse partners). Our model suggests novel avenues for testing the bet-hedging hypothesis of differential susceptibility, including empirical predictions and relevant measures.

Wednesday, 27th: 10:50-11:15, Aula

Emergence of simplicity as a product of cumulative cultural evolution

Masanori Takezawa[1](m.takezawa@let.hokudai.ac.jp), Masaki Suyama[2]
[1]Hokkaido University, [2]Sophia University

Objective

A distinct feature of human culture is its cumulativeness. Knowledge and technology are accumulated through transmission from older generations and reach a high level which cannot be attained by a single individual. As Mesoudi (2011) pointed out, however, culture often becomes increasingly more complex as it is accumulated and it becomes more difficult to copy accumulated knowledge and technology from previous generations. Consequently, increased complexity of culture hinders innovations and eventually stops cumulative cultural evolution. An objective of this study is to show a possibility that culture becomes simpler as it is accumulated so that it can suppress acquisition cost (i.e., keep it easy to be copied from previous generations) and help culture continues to evolve cumulatively.

Methods

We replicated a laboratory experiment of cumulative cultural evolution originally conducted by Caldwell & Millen (2008) with two conditions; group condition (n=7) in which technology of building a tower with cray and spaghetti is transmitted from generation to generation and fixed-pair condition (n=12) in which the same two individuals repeatedly built a tower for the same number of times as in Group condition. Thus, in the fixed-pair condition, technology can be improved without transmission.

Results

In both conditions, technology evolved similarly; height of towers linearly increased and there were no significant differences between the conditions. However, shape of towers became more similar only in the group condition. Such a convergence of technology did not occur in the pair condition. Analysis of tower structure further revealed that towers in group condition became simpler as the technology accumulated while such emergence of simplicity did not occur in the pair condition.

Conclusions

We found that cultural transmission works as selection pressure to simplify accumulated culture. Such changes of culture will suppress cultural acquisition cost and help cumulative cultural evolution continues.

Wednesday, 27th: 11:15-11:40, Aula
The evolution of social group essentialism and intergroup prejudice

Cristina Moya[1] (cristina.moya@lshtm.ac.uk), Scelza Brooke[2]
[1]London School of Hygiene and Tropical Medicine, [2]UCLA

Treating social group membership as stable and inherent is often associated with intergroup prejudice and perceptions of large intergroup differences. However, the functional evolutionary origins of such beliefs, and their causal relation to prejudice, are unclear. First we discuss several possible evolutionary origins of beliefs about group stability, and preferences for group endogamy rooted in reasoning about species, primate troupes, or culturally distinct ethnic groups. Second, we discuss the functional relation (or independence) of these beliefs. Finally, we test these hypotheses by examining the effect of cultural distance between social groups on these beliefs among Himba pastoralists in Namibia, who have undergone recent ethnogenesis. In a vignette experiment we asked participants whether an individual born in one group who moves to another group would retain her original group membership or acquire the new group's membership and cultural characteristics. We varied whether the vignette was about a migration involving a culturally and linguistically related group (the Herero) or a more distant, unrelated group (the Damara). Participants seldom thought of either group boundary as stable, but if anything, treated the boundary between unrelated groups (Damara-Himba) as more fluid than the boundary between related groups. This was despite the fact that nearly all participants expressed more anti-Damara than anti-Herero sentiment with respect to exogamy and despite the larger cultural and historical difference between the Damara and Himba.

Wednesday, 27th: 11:40-12:05, Aula

The Outcome of Merging Cultures

Fredrik Jansson[1] (fredrik.jansson@intercult.su.se), Pontus Strimling[1], Mikael Parkvall[2]
[1]Centre for the Study of Cultural Evolution, [2]Stockholm University

Objectives

A common cultural exchange is when several groups of people each have a set of traits, and where the groups, due to extensive communication, form a common culture holding only one of the traits for each domain. Such situations occur when groups of people unite and develop common norms or language. What is required for people to converge on one trait and what will then be the resulting culture?

Methods

We have developed, analysed and empirically tested a mathematical model of this process, where agents have preferences for different traits and communicate in pairwise interactions where the efficiency of transmission may depend on whether the individuals use the same trait in the interaction. The situation applies to several cultural domains, and particularly suited for studying cultural transmission is language. Creoles are languages that evolved due to limited contact between groups of different linguistic descent and whose evolutionary process thus resembles that of our model particularly well. We used exemplary data from the historical demography of Mauritius as input to the model and compared the results to Mauritian Creole.

Results

We have shown analytically that for a common culture to develop, agents must learn more when communication is successful than not, that is, they need a confirmation bias. Under this assumption, we ran simulations and compared the results to Mauritian creole, giving a good fit.

Conclusions

Our model, together with mathematical analysis and simulations, formulates some possible and impossible processes underlying the development of a common culture, and also predicts which traits will win. For our data, the model provides correct predictions taking only demographic considerations, making specific assumptions such as a prestige bias unnecessary. It also shows that with extensive communication, a confirmation bias promotes a common culture rather than polarisation.

Wednesday, 27th: 10:50-11:15, Auditorium

Personal experience and reputation interact in human decisions to help reciprocally

Lucas Molleman[1], Eva van den Broek[2], Martijn Egas[2] (egas@uva.nl)
[1]University of Groningen, [2]University of Amsterdam

Objective
There is ample evidence that human cooperative behaviour towards other individuals is often conditioned on information about previous interactions. This information derives both from personal experience (cf direct reciprocity) and from experience of others, i.e. reputation (cf indirect reciprocity). Direct and indirect reciprocity have been studied separately, but humans often have access to both types of information. Here, we study whether personal experience and reputation interact in human decisions to help reciprocally

Methods
We experimentally investigate information use in a standard repeated helping game. When acting as donor, subjects can condition their decisions to help recipients on both types of information at a small cost to access such information.

Results
We find that information from direct interactions (personal experience) weighs more heavily in decisions to help than information from interactions with others (reputation), and participants tend to react less forgivingly to negative personal experience than to negative reputation. Moreover, effects of personal experience and reputation interact in decisions to help. If a recipient's reputation is positive, the personal experience of the donor has a weak effect on the decision to help, and vice versa. Yet, if the two types of information indicate conflicting signatures of helpfulness, most decisions to help follow personal experience.

Conclusions
To understand the roles of direct and indirect reciprocity in human cooperation, they should be studied in concert, not in isolation.

Wednesday, 27th: 11:15-11:40, Auditorium

Chimpanzees flexibly adjust their behaviour in order to maximize payoffs, not to conform to majorities

Edwin van Leeuwen[1] (edwin.vanleeuwen@mpi.nl), Katherine Cronin[1], Josep Call[1], Daniel Haun[1&2]
[1]Max Planck Institute for Psycholinguistics, [2]Max Planck Institute for Evolutionary Anthropology

Introduction

Chimpanzees have been shown to be adept learners, both individually and socially. Yet, their conservative nature seems to hamper the flexible adoption of superior alternatives, even to the extent that they persist in using entirely ineffective strategies.

Objective

In this study, we investigated chimpanzees' behavioural flexibility in two different conditions under which social animals have been predicted to abandon personal preferences and adopt alternative strategies: i) under influence of majority demonstrations (i.e. conformity), and ii) in the presence of superior reward contingencies (i.e. maximizing payoffs). Unlike previous nonhuman primate studies, this study disentangled the concept of conformity from the tendency to maintain one's first-learned strategy. Methods & Results: Studying captive (n=16) and semi-wild (n=12) chimpanzees in two complementary exchange paradigms, we found that chimpanzees did not abandon their behaviour in order to match the majority, but instead remained faithful to their first-learned strategy, which is indicative of conservatism, not conformity (experiment 1a and 1b). However, the chimpanzees' fidelity for using their first-learned strategy was overridden by an experimental upgrade of the profitability of the alternative strategy (experiment 2).

Conclusions

Contrary to previous findings, these results show that chimpanzees may abandon their familiar behaviour to maximize payoffs, but not to conform to a majority. We interpret our observations in light of evolutionary continuity with human conformity and in terms of chimpanzees' relative weighing of learning heuristics as a function of situation-specific trade-offs.

Wednesday, 27th: 11:40-12:05, Auditorium
The evolutionary origins of selfish and servant leadership

Edward Cartwright[1] (E.J.Cartwright@kent.ac.uk), Anna Stepanova[1]
[1]University of Kent

Objective
We explore the evolutionary origins of different types of leadership and followership. Particular focus is put on the distinction between a selfish leader, someone who exploits power for their own gain, and a servant leader, someone who is willing to sacrifice for the good of others. We know that both types of leader exist, but how can such starkly different types of leadership emerge and coexist?

Methods
To answer this question we use an evolutionary game theory approach. We analyze a dynamic coordination game similar to the battle of the sexes game. Strategies in the game include selfish leadership, servant leadership, and followership. This allows us to draw a clear distinction between different types of leader; a selfish leader, for example, chooses own preferred action while a servant leader chooses the action preferred by the group. We solve for the evolutionary stable strategies, determine the relevant basins of attraction, and characterize the dynamic convergence to equilibrium. This is done for both a two player and three player version of the game. Results: We show that there will be convergence to an evolutionary stable strategy. The evolutionary stable strategy of most interest is one in which selfish and servant leaders, and followers, co-exist. The proportion of each type depends on the parameters of the game. It is shown that selfish and servant leaders complement each other and so it is natural that both types of leader should emerge.

Conclusions
We provide an evolutionary explanation for the existence of distinct types of leader. We also illustrate different factors that may influence the proportion of each type of leader in the population.

Wednesday, 27th: 14:30-14:55, Auditorium

Costly punishment as a strategy for optimising public goods investment

Joanna Bryson (jjb@cs.bath.ac.uk)
University of Bath

Objective

Account for variation by global region in antisocial punishment (ASP – sanctioning those contributing more than the sanctioner) in public goods games (PGG), as described by Herrmann, Thöni & Gächter (2008).

Method

We examined strategies expressed by Herrmann et al's subjects. We also looked at the impact of having a group member invest all (Cooperator) or none (Freerider) of their allotment in the PG. We created evolutionary ABM exploring the coherence of our hypotheses.

Results

In Herrmann et al's data, rates of ASP vary more than those of altruistic punishment (AP). Variation across subject pools in the proportion of Cooperators is much greater than in the proportion of Defectors. We found no significant correlation between AP and the proportion of Cooperators, AP and Freeriders, and ASP and Freeriders, but a strong anticorrelation between ASP and the proportion of Cooperators ($r = -0.62$, $p < 0.01$). Many more individuals employed both AP and ASP than ASP exclusively. Exclusive AP and complete abstention from punishment were widely-used strategies. When an individual is punished, they are less likely to invest the same amount in the PG in the next round. ASP results in random variation, whereas AP results in increased public goods investment (PGI). Our models show that when PGI is subject to selection in regularly varying contexts, populations form with peaks of similar PGI strategies across a range of values extending both above and below optimal PGI levels.

Conclusions

We believe that overall levels of PGI are generated by populations composed of multiple strategies, including over and under investment (cf. Maclean et al 2010). Shifting between strategies as a result of individual experience may allow agile tracking of changing economic landscapes. Punishment may be a social mechanism accelerating such tracking. We are currently studying its impact with further simulations.

Wednesday, 27th: 14:55-15:20, Auditorium

Divergent selection but no genetic conflict over female and male timing and rate of reproduction in a human population

Elisabeth Bolund[1] (e.bolund@sheffield.ac.uk), Sandra Bouwhuis[2], Jenni Pettay[3], Virpi Lummaa[1]
[1]University of Sheffield, [2]University of Oxford, [3]University of Turku

Objective
The sexes often have different phenotypic optima for important life-history traits and because they share much of their genome, this can lead to a conflict over trait expression. In mammals, the obligate costs of reproduction are higher for females, making reproductive timing and rate especially liable to conflict between the sexes. While studies from wild vertebrate populations show support for such sexual conflict, it remains unexplored in humans.

Methods
We used demographic data with up to 12 generations of pedigree depth collected from church registers of 7 parishes in pre-industrial Finland to estimate sexual conflict over age at first and last reproduction, reproductive lifespan and reproductive rate.

Results
We found that the phenotypic selection gradients differed between the sexes. For age at first and last reproduction and reproductive lifespan, the relationships with fitness (number of grandchildren) tended to be nonlinear in women, suggesting an intermediate optimum value, while they were linear in men. Both sexes showed a linear decrease in fitness with increasing reproductive rate. We next established significant heritabilites in both sexes for all traits. All traits, except reproductive rate, showed strongly positive intersexual genetic correlations and were strongly genetically correlated with fitness in both sexes. Moreover, the genetic correlations with fitness were almost identical in men and women. The intersexual correlation and the correlation with fitness were both weaker for reproductive rate, but again similar between the sexes.

Conclusions
These findings illustrate that apparent sexual conflict at the phenotypic level is not necessarily indicative of an underlying genetic conflict and further emphasize the need for incorporating a genetic perspective into studies of human life-history evolution.

Wednesday, 27th: 15:50–16:15, Auditorium

Group competition eliminates indiscriminate punishment and promotes cooperation

Aljaz Ule[1] (a.ule@uva.nl), Arthur Schram[1], Arno Riedl[2], Timothy Cason[3]
[1]University of Amsterdam, [2]Maastricht University, [3]Purdue University

Objective
This paper investigates how competition across groups affects the use of intragroup punishment and the resulting dynamics of intragroup cooperation. This is important because little is known about the dynamics of punishment and cooperation under group competition.

Methods
Groups of six human subjects play 100 periods of a laboratory indirect reciprocity game. In each period the subjects are randomly matched into three donor–recipient pairs. The donor can donate, punish (at a cost) or snub her recipient after observing her three last donation decisions. In the 'Isolated Groups' (IG) treatment the final point earnings of the six group members are ranked at the end of the experimental session and an individual's monetary payment depends on her within-group rank. In the 'Competing Groups' (CG) treatment two groups play the game in parallel. The point earnings of all twelve individuals from both groups are ranked and the individual's monetary payoff depends on her across-group rank. Hence GC adds competition with outside groups to the within-group interaction. We study the dynamics of punishment and cooperation and the strategies that individuals apply.

Results
Compared to IG, CG substantially increases cooperation (35% vs. 15%) and decreases punishment (12% vs. 32%). In IG punishment is typically indiscriminate and positive reciprocation is rare. In CG, punishment is mainly directed towards free-riders but many more subjects reward by donating to frequent donors. In CG, individuals also much more often consult recipient's reputation and apply image scoring or standing strategies.

Conclusions
Our results show that group competition increases cooperation and reduces intragroup conflict. This suggests that competition between human groups or tribes may have played a key role in the evolution of strong reciprocity, such as costly punishment of norm violators.

Wednesday, 27th: 16:15-16:40, Auditorium
Pupillary contagion influences social decisions

Mariska Kret[1] (m.e.kret@uva.nl), Jolien van Breen[2], Agneta Fischer[1], Carsten de Dreu[1]
[1]University of Amsterdam, [2]University of Groningen

Following each other's gaze and attending to pupil size allows humans to communicate information about the immediate environment. Building on these findings I will present different behavioral and pupillometry experiments investigating "pupillary contagion" - the synchronization of pupil size with an interaction target. I will argue that pupillary contagion has adaptive value, e.g., to promote shared understanding and coordination, because it emerges within, but not across species. This can be observed when humans synchronize their pupils specifically with other humans, but not with chimpanzees (P. troglodytes). Moreover, I will argue that pupil synchronization is uniquely human, as it does not occur in our closest relative, the chimpanzee. I will discuss the idea that eye white (which is not visible in chimpanzees) may be an adaptation to enhance swift communication of arousal (gaze following & pupillary contagion) within groups. Communication of arousal via eye signals can only have evolved in groups where individuals trusted each other and cooperated, not in groups where others would abuse such information for their own benefit. This is demonstrated in studies on the effects of pupillary contagion on trust and deception, where pupillary contagion induced trust and reduced the likelihood of deception.

Wednesday, 27th: 16:40-17:05, Auditorium
Morphological versus perceptual measures of masculinity

Lisa DeBruine[1] (lisa.debruine@glasgow.ac.uk), Daniel Re[2], David Perrett[2], Corey Fincher[1], Benedict Jones[1]
[1]University of Glasgow, [2]University of St Andrews

Objective
The extent to which women emphasise cues of masculinity when assessing potential mates has recently been questioned in light of negative findings relying on the new technique of measuring "morphological masculinity" using a discriminant score based on sex differences in 2D face shape. Here, we attempt to validate the use of this technique by comparing discriminant scores with perceptual ratings of masculinity/femininity and dominance.

Methods
Following published techniques, we calculated discriminant scores for two samples of 100 faces (50 male and 50 female). Both unmanipulated versions and colour-standardised versions of these faces in which colour and texture cues had been replaced with the average same-sex colour and texture (leaving only 2D shape cues) were rated for masculinity/femininity. The unmanipulated versions were also rated for dominance, which is strongly and consistently linked to both manipulations and ratings of male facial masculinity. Each combination of sample, face sex, stimulus type and rating was completed by a separate group of raters.

Results
In both samples, the discriminant score failed to significantly correlate with human perceptual ratings of the masculinity of either the men's unmanipulated faces or colour-standardised versions. However, the discriminant score did positively correlate with masculinity ratings of the women's original faces and colour-standardised versions. Additionally, the discriminant score did not predict dominance ratings of male faces in either sample.

Conclusion
Discriminant scores failed to consistently predict human perceptual ratings of the masculinity or dominance of male faces, even when colour and texture cues were standardised so that only shape cues were assessed. However, discriminant scores did predict human perceptual ratings of the femininity of female faces. The results indicate that using discriminant scores may not be suitable for assessing male facial masculinity, which may alter the interpretation of previously reported findings.

Wednesday, 27th: 17:05-17:30, Auditorium

Kin and Fertility in Indonesia: How Do Different Measures of Kin Availability Affect Fertility Outcomes?

Kristin Snopkowski[1](kristin.snopkowski@lshtm.ac.uk), Rebecca Sear[1]
[1]London School of Hygiene and Tropical Medicine

Objectives
Evidence has shown that family members influence fertility outcomes. A problem with this literature, however, is that kin presence is measured in many different ways, depending on data availability, which limits the comparability of previous work. In many studies the survival status of family members is used as a proxy for kin availability. Others have measured kin availability by co-residence with kin, particularly postnuptial residence. This may be biased if individuals who live with kin after marriage are systematically different from those who do not. Here we use a dataset that allows us to compare the effects of different measures of kin availability on fertility, so we can determine whether the results are similar using different measures of kin availability.

Methods
Data are derived from four waves of the Indonesia Family Life Survey (IFLS). Multiple regression analyses were used to predict the status of parents at each wave on fertility outcomes. The progression to a birth between waves was analyzed using parents status at the beginning of the wave. Frequency of contact and distance from kin were used as indicators of kin's helping ability or likelihood of influence.

Results
Women living with their mother-in-law's gave birth to more children and have more living children. Women living with their parents have fewer children (particularly in low-income households)

Conclusion
Our results show that fertility outcomes are quite different for individuals living with kin versus those who have surviving kin.

Poster list click number to open poster (if available)

1. **Late bird catches the girl? Relationship between eveningness chronotype, Dark Triad of personality, and socio-sexual orientation**
   Amy Jones[1] (10100260@hope.ac.uk), Minna Lyons[1]
   [1]Liverpool Hope University

2. **Sex differences in social information use**
   Catharine Cross[1] (cpc2@st-andrews.ac.uk), Gillian Brown[1], Thomas Morgan, Kevin Laland[1]
   [1]University of St Andrews

3. **Switching on and off the pill: Outcomes for relationship jealousy**
   Kelly D. Cobey[1] (k.d.cobey@rug.nl), Craig S. Roberts[2], Abraham P Buunk[1]
   [1]University of Groningen, [2]University of Stirling

4. **Digit ratio and risk taking in post-menopausal Finnish women**
   Minna Lyons[1] (lyonsm2@hope.ac.uk), Samuli Helle[2]
   [1]Liverpool Hope University, [2]University of Turku

5. **Swarm Intelligence in a multi-armed bandit task with word-of-mouth**
   Wataru Toyokawa[1] (toyokawa@lynx.let.hokudai.ac.jp), Hye-rin Kim[1], Tatsuya Kameda[1]
   [1]Hokkaido University

6. **The made-up lass is of higher class: Sex differences in judgements of dominance and prestige, but not attractiveness, for women with and without make-up.**
   Viktoria R. Mileva[1](v.r.mileva@stir.ac.uk), Alex Jones[2], Richard Russell[3], Anthony Little[1]
   [1]University of Stirling, [2]University of Bangor, [3]Gettysburg College

7. **The attractiveness of humour types in personal advertisements: Affiliative and aggressive humour are differentially preferred in long-term versus short-term partners.**
   Mary Louise Cowan[1](m.l.cowan@stir.ac.uk), Anthony C. Little[1]
   [1]University of Stirling

8. **The unusual box test – children's individual learning through object exploration and divergent thinking**
Simone Bijvoet-van den Berg[1](c.j.bijvoet@stir.ac.uk), Elena Hoicka[2]
[1]University of Stirling, [2]University of Sheffield

9. **Serial killers, spiders and cybersex: Social and survival information bias in the transmission of urban legends.**
Joseph Stubbersfield (j.m.stubbersfield@durham.ac.uk)
University of Durham

10. **The function of lateralization**
Nele Zickert[1](nelezickert@gmail.com), Tess Beking[1], Ton Groothuis[1], Reint Geuze[1]
[1]University of Groningen

11. **A hunger for justice: Food-stress and moral cognition in a rural Ethiopian population**
Isabel Scott[1] (isabel.scott@brunel.ac.uk), Burka Tessema[2], Mhairi Gibson[3]
[1]Brunel University, [2]Addis Ababa University, [3]University of Bristol

12. **What does arousal look like? Temperature and color changes in the face**
Amanda Hahn[1&2](a.hahn@psy.gla.ac.uk), Carmen Lefevre[2], David Perrett[2]
[1]University of Glasgow, [2]University of St Andrews

13. **Gender differences in the computational processing of sibling detection cues. A psychophysiological study of incest aversion.**
Delphine de Smet[1] (delphine.desmet@ugent.be), Linda van Speybroeck[1], Jan Verplaetse[1]
[1]Ghent University

14. **Machiavellianism, competition and self-disclosure in friendship**
Loren Abell[1] (labell@uclan.ac.uk), Gayle Brewer[1], Minna Lyons[2],
[1]University of Central Lancashire, [2]Liverpool Hope University

15. **Assortative mating for weight is not due to assortative preferences**
Claire I. Fisher[1] (claire.fisher@glasgow.ac.uk), Corey L. Fincher[1], Anthony C. Little[2], Lisa M. DeBruine[1], Benedict C. Jones[1]
[1]University of Glasgow, [2]University of Stirling

16. Testosterone and cortisol release among Spanish soccer fans watching the 2010 world cup final

Leander van der Meij[1] (L.van.der.Meij@vu.nl), Mercedes Almela[2], Vanesa Hidalgo[2], Carolina Villada[2], Paul A.M. van Lange[1], Alicia Salvador[2]
[1]VU University Amsterdam, [2]University of Valencia

17. Ordinary heroes? An evolutionary analysis of war hero Biographies

Hannes Rusch (hannes.rusch@uni-giessen.de)
University of Giessen

18. Milgram revisited: Contextual influences on imitative behavior

Julie Coultas[1&2] (julie.coultas@gmail.com), Kimmo Eriksson[1],
[1]Stockholm University, [2]University of Sussex

19. Female condition affects perception of male body attractiveness

Vít Třebický[1] (vit.trebicky@gmail.com), Jan Havlíček[1]
[1]Charles University

20. Different mechanisms for disgust

Dieneke Hubbeling (d.hubbeling@btinternet.com)
University of East London

21. The cultural evolution of ineffective medicine

Mícheál de Barra[1] (mdebarra@gmail.com), Pontus Strimling[1], Kimmo Eriksson[1]
[1]University of Stockholm

22. Machiavellianism and emotional intelligence

Linda Szijjarto (lindaszijjarto@gmail.com)
University of Pécs,

23. Cultural population differences, cumulative culture, and the difference between human and nonhuman culture

Marius Kempe[1] (marius.kempe@me.com), Alex Mesoudi[1], Stephen Lycett[2]
[1]Durham University, [2]University of Kent

24. **Men's, but not women's, sociosexual orientation predicts couples' perceptions of sexually dimorphic cues in own-sex faces**

Michal Kandrik[1] (2054031K@student.gla.ac.uk), Corey L. Fincher[1], Benedict C. Jones[1], Lisa M. DeBruine[1]
[1]University of Glasgow

25. **Facial width-to-height ratio predicts dominance and status in the primate face**.

Carmen E. Lefevre[1] (cel37@st-andrews.ac.uk), Vanessa A.D. Wilson[2], Blake F. Morton[3], Sarah Brosnan[4], Annika Paukner[5], Tim C.D. Bates[2]
[1]University of St. Andrews, [2]University of Edinburgh, [3]University of Stirling, [4]Georgia State University, [5]Eunice Kennedy Shriver National Institute of Child Health and Human Development, National Institutes

26. **Red is not a proxy signal for female genitalia in humans**

Sarah E. Johns[1] (s.e.johns@kent.ac.uk), Lucy A. Hargrave[1], Nicholas E. Newton-Fisher[1]
[1]University of Kent

27. **Female mate retention, sexual orientation and gender identity**

Gayle Brewer[1] (GBrewer@UCLan.ac.uk), Victoria Hamilton[1]
[1]University of Central Lancashire

28. **Lower intelligence in *Toxoplasma gondii* infected men**

Veronika Chvátalová[1] (veronikachvatalova@seznam.cz), Jaroslav Flegr[1]
[1]Charles University

29. **The human body odour composites are not more attractive than individual samples**

Jitka Fialová[1] (jitka.fialova@fhs.cuni.cz), Agnieszka Sorokowska[2], Craig S. Roberts[3], Lydie Kubicová[1], Jan Havlíček[1]
[1]Charles University, [2]University of Wroclaw, [3]University of Stirling,

30. **Religious practices, personality and spirituality: which one better predicts generosity?**

Tiago Soares Bortolini[1] (tbortolini@gmail.com), Karoline Vieira Dantas[1], Nayara Lima Soares[1], Thamires Pinto Soares[1], Tadashi Hattori Wallisen[1], Emilia Yamamoto Maria[1]
[1]Universidade Federal do Rio Grande do Nort

31. **In search of legal memes**
Derk Venema (d.venema@jur.ru.nl)
Radboud University Nijmegen

32. **Kin influences on fertility: a theoretical framework tested with a meta-analysis of the literature**
Rebecca Sear[1] (rebecca.sear@lshtm.ac.uk), Cristina Moya[1], Paul Mathews[2]
[1]London School of Hygiene and Tropical Medicine, [2]University of Essex

33. **Aggression, mate value and the preference for faces of fertile females**
Cora Bobst[1] (cora.bobst@psy.unibe.ch), Janek S. Lobmaier[1]
[1]University of Bern

34. **Body attractiveness, nasopharyngeal flora and immunocompetence**
Boguslaw Pawlowski[1] (bogus@antropo.uni.wroc.pl), Judyta Nowak[1], Barbara Borkowska[1], Zuzanna Drulis-Kawa[2]
[1]University of Wroclaw, [2]University of Wroclaw

35. **Private gain and the public good: The potential role of dominance in the evolution of third-party punishment**
David, S. Gordon[1] (dsg204@exeter.ac.uk), Stephen E. G. Lea[1], Joah, R. Madden[1]
[1]University of Exeter

36. **Self-blindness in humans as prerequisite for the evolution of advanced intelligence. (Towards “Amathology” or the science of ignorance)**
Popko P. van der Molen (popko@sicirec.org)
R.U.G.

37. **Who is a clever girl then?! Investigating reputation model-based transmission biases**
Lara Wood[1] (l.a.n.wood@dur.ac.uk), Rachel Kendal[1], Emma Flynn[1]
[1]Durham University

38. Grandparental Investment, parents' fertility, and child development in the UK

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[1]University of Helsinki

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Yo Nakawake[1] (wake0510@gmail.com), Masanori Takezawa[2]
[1]Sophia University, [2]Hokkaido University

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Sebastian Grueneisen[1] (sebastian_grueneisen@eva.mpg.de), Emily Wyman[1], Michael Tomasello[1]
[1]Max Planck Institute for Evolutionary Anthropology

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[1]University of Bristol

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Bronwyn Tarr (bronwyn.tarr@psy.ox.ac.uk)
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Katrin Schaefer[1] (katrin.schaefer@univie.ac.at), Sonja Windhager[2], Philipp Mitteroecker[1]
[1]University of Vienna, [2]Konrad Lorenz Institute for Evolution and Cognition Research

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[1]VU University Amsterdam

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Rei Shimoda (rei.shimoda@durham.ac.uk)
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[1]Utrecht University

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Sue Aitken[1] (aitkens@hope.ac.uk), Tom Platt[1]
[1]Liverpool Hope University

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[1]Queensland Institute of Medical Research, [2]University of Queensland, [3]VU University Amsterdam

96. Menstrual cycle effects on attitudes towards romantic kissing

Rafael Wlodarski (rafael.wlodarski@psy.ox.ac.uk), Robin I.M. Dunbar
University of Oxford

Open Access Panel

"The Future of Publishing in Evolutionary Sciences" plenary round-table panel discussion at EHBEA conference, Monday 25 March 2013, 5.30-7.00 pm, Auditorium VU main building

Should open access publishing be the norm in evolutionary sciences? What are the implications of open access publishing for the way scientists and journals do their work? What are the financial costs of open access publishing and who should bear these costs? Should the scientists pay to get their work published, the readers of the articles, or perhaps universities or the government? These are some of the questions that will be addressed at this unique panel discussion on the future of publishing in the evolutionary sciences. There is plenty of room for audience participation so please join us to make this a lively debate! This session is sponsored by the Dutch Science foundation (NWO)

Panel members:

Prof. Kristen Hawkes (University of Utah)

Prof. Peter Nijkamp (VU University-professor and former chair of NWO)

Fiona Routley (Springer Publisher)

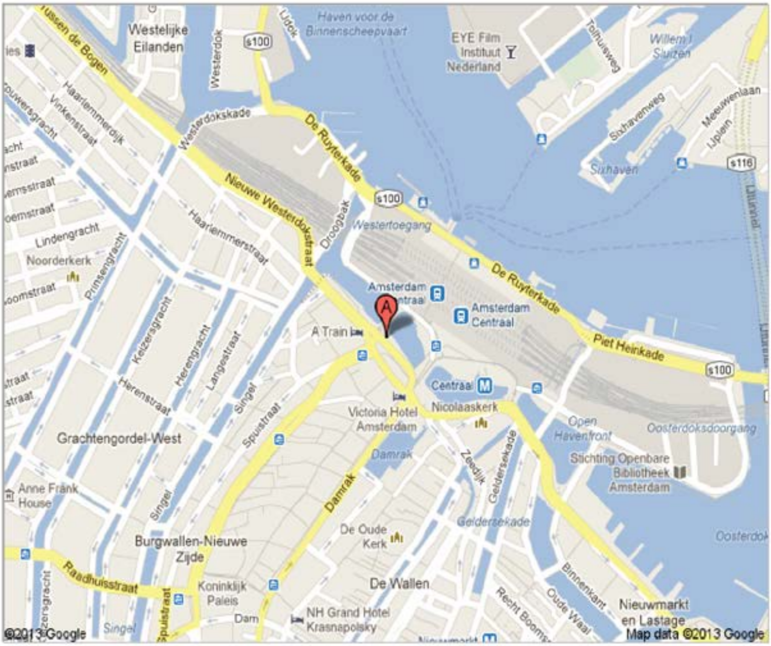
Prof. Rebecca Sear (London School of Hygiene and Tropical Medicine)

Boat ride

The Amsterdam canals celebrate their 400th anniversary this year. We have arranged for you to experience this part of our history by night. The boat will pass by many of the major sights Amsterdam has to offer, such as the gorgeous, distinctly Dutch houses lining the inner-city canals, the gorgeous Westerkerk (Western Church), and Amsterdam's famous Centraal Station. This is a unique and relaxing way to see what Amsterdam has to offer!

On Monday, March 25th, two boats will leave from the "Canal" dock (Canal is the name of the company) in front of Amsterdam Central Station at **9.20 pm** (boat leaves with or without you at 9.30pm). Please aim to arrive early; we don't want anyone to miss the boat! There is a complimentary drink included with the price of your boat ticket. The boat will return at approximately 10.30pm.

Boat Ride: Directions to the loading dock



The address is the 'Holland International Rondvaart' dock directly across from Amsterdam Central Station. Address: Prins Hendrikkade 33a.

From the conference venue, you can take the bus (170), metro (51), or tram (5) to central station. The trip should take around 25-30 minutes. Please ask the registration desk if you need further information on transportation.

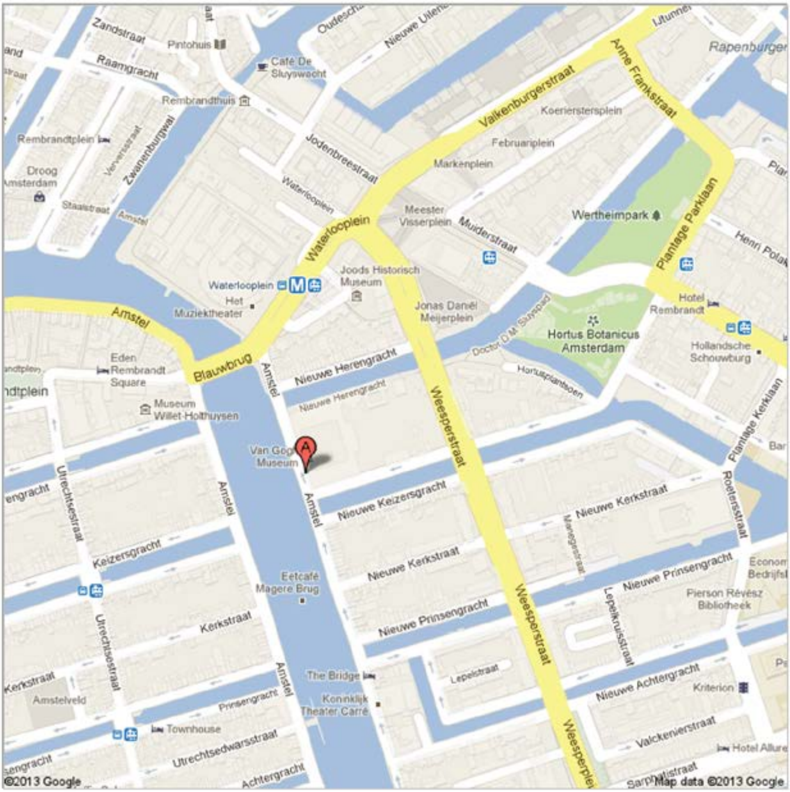
Conference Dinner

We have booked a wonderful location in the center of Amsterdam for the conference dinner: Restaurant Neva, located at the Hermitage Amsterdam. Chef Ricardo van Ede, who received a Michelin star when he was 21, and his team will treat you to a buffet with a variety of dishes from the French-international kitchen.

The building in which Hermitage Amsterdam is housed, the Amstelhof, served as a home for the elderly for 324 years. At the close of the twentieth century, it became apparent that the Amstelhof care facility no longer met contemporary standards, and a new use was sought for the building. Since June 2009, the site has been home to Hermitage Amsterdam, which is a branch of the Hermitage Museum of Saint Petersburg, Russia. The Hermitage is one of Amsterdam's most impressive museums, with permanent collections detailing Dutch-Russian relationships, and temporary exhibitions displaying some of the world's greatest artists (e.g., a current Van Gogh exhibition). You can learn more about the restaurant housed in the Hermitage here:

http://www.neva.nl

Public transport from conference:
Take Metro 51 toward Centraal Station. Stop at the Waterlooplein metro stop, and take the Nieuwe Herengracht exit (walk along the Nieuwe Herengracht in the direction of the Amstel river) or the Stadhuis exit (walk to the bridge over Amstel river and turn left along the river – don't cross the Amstel!).



Location A in the map is the venue for the conference dinner. Please note: Google Maps calls this the ‘Van Gogh Museum’, as the Van Gogh exhibit is currently at Hermitage.

The address is Amstel 51, 1018 DR Amsterdam.

Useful information

Emergency
European emergency call is 112. If you need to reach somebody at the conference you can contact Conference Coordination +31 (0)20 598 5942. Or: Thomas Pollet: +31 (0)6 81705536.

Campus security contact number
Emergency number: +31 (0)20 598 5854.

Internet Access
The VU University provides free wireless internet access for those who are part of 'Eduroam'. To connect please select the network named "Eduroam". There is no username or password required.
You can also get a wifi code from the registration desk, if you do not have an "Eduroam" account. If you have problems, please visit the registration desk.

Public transport
Amsterdam offers an extensive travel network using metro, tram and bus. The closest trams are 5, 16 and 24. The closest metro is 51.

The public transport card ('OV chip card') allows use of public transport in the city. It can be obtained at train stations and most metro stations.

For metro, tram see: http://en.gvb.nl/pages/home.aspx
For trains see: http://www.ns.nl/en/travellers/home

Contact the registration desk, if you need directions.

Taxis
TCA (Taxi Central Amsterdam) is the official taxi company for the city Amsterdam.
More information can be found on:
www.tcataxi.nl/en
To order a taxi you can call: +31(0)20 777 7777

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